闫一慧

(+86)180-3518-1760 | 1753199938@qq.com | github.com/ElenaHUI

教育经历

华东师范大学 | 软件工程学院 | 软件工程专业 2024.09 —至今

研究方向：边缘计算（MEC）、物联网（IoT）、无人机（UAV）与深度学习的融合。

西安电子科技大学 | 计算机科学与技术学院 | 软件工程专业 2020.09 —2024.06

均分：88.1，GPA：3.8/4.0，专业排名：21/332（软件工程）| 4/61（网络与通信方向）

主修课程：数据结构（99）、计算机网络与通信（91）、编译原理（97）、算法（99）、高等数学（90）等。

连续四年获得校级奖学金、全国大学生数学建模大赛陕西省一等奖、优秀共青团员、优秀毕业生

项目经历

华为技术有限公司 | AI 工程师实习生 |*2025.07 —2025.10,* 上海

参与大模型测试平台（云探）项目开发，在云端服务器利部署模型，通过 MCP 框架搭建智能体实现压测任务自动生成与调度，并开发功耗监控组件进行数据采集、分析与可视化，同时完善平台后端功能，**累计贡献代码 2000+**。

- **大模型部署：**使用云端服务器并基于 Ascend-vLLM 框架完成基于 Llama 和 Qwen 等模型的推理服务非分离部署与训练服务部署，并利用内部平台进行性能压测，获得包括 QPS、TTFT 在内的多个性能压测指标；
- **性能采集组件设计：**搭建服务器 + Prometheus + Grafana 监控测试链路，并基于该链路独立开发功耗采集组件（A3-Power），实现多个 NPU 节点中功耗、内存占用率、NPU 温度等 9 个指标的可视化与自动化增量分析；
- **MCP 智能体实现：**基于 FastMCP 框架搭建本地 MCP server 并利用内部灵雀平台 (Dify) 进行智能体搭建，实现基于用例描述的压测任务的生成、执行及跨网部署，从而实现测试提效，**多任务执行效率由 4 小时提升至 30 秒**；
- **Java 后端平台优化：**在 API 层、DAO 层、Common 层完整实现工作逻辑；完成与前端的接口联调；独立完成 Mapper 层测试用例，确保权限与查询功能在全链路上的正确性和稳定性。

"小神探"设备维修管理系统 |*2022.11 —2023.06*

- 主导开发基于 **SpringBoot + Vue3** 的前后端分离系统，支持设备信息存储、报修、保养、维修全流程管理；
- 负责系统架构设计与接口规范制定，提高运维效率与系统可扩展性。

玻璃成分智能分析与分类系统 |*2021.08 —2021.12*

- 利用 **Fisher 判别法**对不同玻璃成分进行分类；
- 使用 **模糊聚类算法**进行亚类划分，实现风化前后样品的差异分析。

科研经历

边缘计算与无人机传输优化方向 | 指导教师：肖波教授 | 2024.10 **—至今**

[Submitted] *Yihui Yan, Pengfei Ren, et al., MAO-LCR: A Multi-Agent Learning Approach for Joint Task Offloading and Cache Replacement in Edge Computing，IEEE WCNC 2026*

提出一种后悔感知的多智能体强化学习方法，对任务卸载与缓存决策进行优化，降低时延与能耗并提升缓存命中率。

[Under Review] *Yihui Yan, Huimin Cao, et al., Multi-Agent Proximal Policy Optimization for Joint Task Offloading and Trajectory Control in NOMA-based Multi-UAV MEC Networks, IEEE Internet of Things Journal(IoT)*

提出一种多智能体强化学习的联合任务卸载与轨迹优化算法，在动态 NOMA-MEC 网络中最小化系统时延与能耗。

计算机视觉方向 | 指导教师：张亮教授 | 2022.02 —2023.06

[Accepted] *Yihui Yan: AlexViT: Novel Diabetic Retinopathy Image Classification, IEEE ICETCI 2023*

结合 ViT (Vision transformer) 的全局注意力机制与 AlexNet 的局部特征提取能力，设计轻量级混合结构的视网膜病变分类模型，显著压缩参数量（45%）并提升 Kappa 值（3.7 %），在 EyePACS 与 Messidor 数据集上取得领先性能。

语言与技能

- **语言：**CET-4（587），CET-6（530）
- **编程：**Java, Python, C++ ；
- **技能：**熟悉 Git 管理流程; LaTeX、Markdown 等文本撰写语言; Simulink 仿真, Arduino, Matlab 等工具

学生工作

中共党员、西安电子科技大学校团委文化部骨干成员、华东师范大学研会人力资源部部长、班级素拓委员。

个人总结

在校成绩优异，具备扎实的计算机与人工智能理论基础；自驱力强，能够独立快速学习并应用新技术；具备科研与工程结合能力，能够将前沿技术在实际系统中落地；具备项目实践经验，能够在团队中高效协作完成所负责功能。

Yihui Yan

(+86)180-3518-1760 | 1753199938@qq.com | github.com/ElenaHUI

Education

East China Normal University |School of Software Engineering *2024.09 —Present*

Research interests: Integration of Mobile Edge Computing (MEC), Internet of Things (IoT), and Deep Learning.

Xidian University |School of Computer Science and Technology *2020.09 —2024.06*

GPA: 3.8/4.0 | Ranking: 21/332 (Software Engineering) & 4/61 (Network Direction)

Main Courses: Data Structure (99), Computer Network and Communication (91), Compiler Principles (97), Algorithms (99), Advanced Mathematics (90), etc.

Awarded university scholarships for four consecutive years; First Prize in Shaanxi Province Mathematical Modeling Contest; Excellent League Member; Outstanding Graduate.

Project Experience

Huawei Technologies Co., Ltd. | AI Engineer Intern |*2025.07 —2025.10, Shanghai*

Participated in the development of the large model testing platform (**Yuntan**), deploying models on cloud servers, building intelligent benchmarking agents via MCP framework for automated task generation and scheduling, and developing power monitoring components for data collection, analysis, and visualization. Also optimized backend functionalities, **contributing over 2000 lines of code.**

- **Model Deployment:** Deployed inference and training services for Llama and Qwen models on cloud servers based on the **Ascend-vLLM** framework, and performed performance benchmarking to obtain key metrics such as **QPS** and **TTFT**.
- **Performance Monitoring Component:** Built a monitoring pipeline using **Prometheus + Grafana**, and independently developed the **A3-Power** component to visualize and analyze **9 metrics** (power consumption, memory usage, NPU temperature, etc.) across multiple NPU nodes with automated incremental analysis.
- **MCP Agent Development:** Constructed a custom MCP server based on **FastMCP** and integrated with the internal **Lingque (Dify)** platform to achieve automatic generation, scheduling, and cross-network execution of benchmarking tasks, improving testing efficiency—**reducing task execution time from 4 hours to 30 seconds.**
- **Java Backend Optimization:** Implemented complete logic across API, DAO, Common, Mapper, and MySQL layers; completed front–back-end API integration and Mapper-level unit tests to ensure system stability and correctness of permission and query functions.

"Little Detective" Device Maintenance Management System |*2022.11 —2023.06*

- Led the development of a **SpringBoot + Vue3** based front–back-end separated system supporting device registration, maintenance requests, and repair management;
- Designed the overall architecture and API specifications, improving system scalability and maintenance efficiency.

Intelligent Glass Composition Analysis and Classification System |*2021.08 —2021.12*

- Applied the **Fisher Discriminant Method** for classification of different glass compositions;
- Used **Fuzzy Clustering Algorithms** to identify subcategories and analyze variations between pre- and post-weathered samples.

Research Experience

Edge Computing and UAV Research | Supervisor: Prof. Bo Xiao |*2024.10 —Present*

[**Submitted**] *Yihui Yan, Pengfei Ren, et al., MAO-LCR: A Multi-Agent Learning Approach for Joint Task Offloading and Cache Replacement in Edge Computing, IEEE WCNC 2026*

Proposed a regret-aware multi-agent reinforcement learning framework to jointly optimize task offloading and caching strategies, effectively reducing latency and energy consumption while improving cache hit rate.

[**Under Review**] *Yihui Yan, Huimin Cao, et al., Multi-Agent Proximal Policy Optimization for Joint Task Offloading and Trajectory Control in NOMA-based Multi-UAV MEC Networks, IEEE Internet of Things Journal (IoTJ)*

Developed a multi-agent reinforcement learning approach for joint task offloading and trajectory optimization in dynamic NOMA-MEC networks to minimize system delay and energy consumption.

Computer Vision Research | Supervisor: Prof. Liang Zhang | *2022.02 —2023.06*

[**Accepted**] *Yihui Yan: AlexViT: Novel Diabetic Retinopathy Image Classification, IEEE ICETCI 2023*

Proposed a lightweight hybrid model combining the global attention of **Vision Transformer (ViT)** with the local feature extraction of **AlexNet**. The model introduces a convolutional embedding and attention fusion mechanism, achieving a **45% parameter reduction** and a **3.7% improvement in Kappa score**, attaining state-of-the-art performance on EyePACS and Messidor datasets.

Languages & Skills

- **Languages:** CET-4 (587), CET-6 (530)
- **Programming:** Java, Python, C++
- **Technical Tools:** Git workflow, LaTeX, Markdown, Simulink, Arduino, Matlab

Student Activities

Communist Party member; Core member of the Cultural Department, Youth League Committee of Xidian University; Director of Human Resources Department, Graduate Student Union, ECNU; Class Development Committee member.

Personal Summary

Outstanding academic performance with a strong foundation in computer science and artificial intelligence. Highly self-motivated, able to independently learn and apply new technologies. Skilled in bridging research and engineering—capable of deploying cutting-edge algorithms into real-world systems. Experienced in project implementation, emphasizing teamwork, reliability, and practical impact.